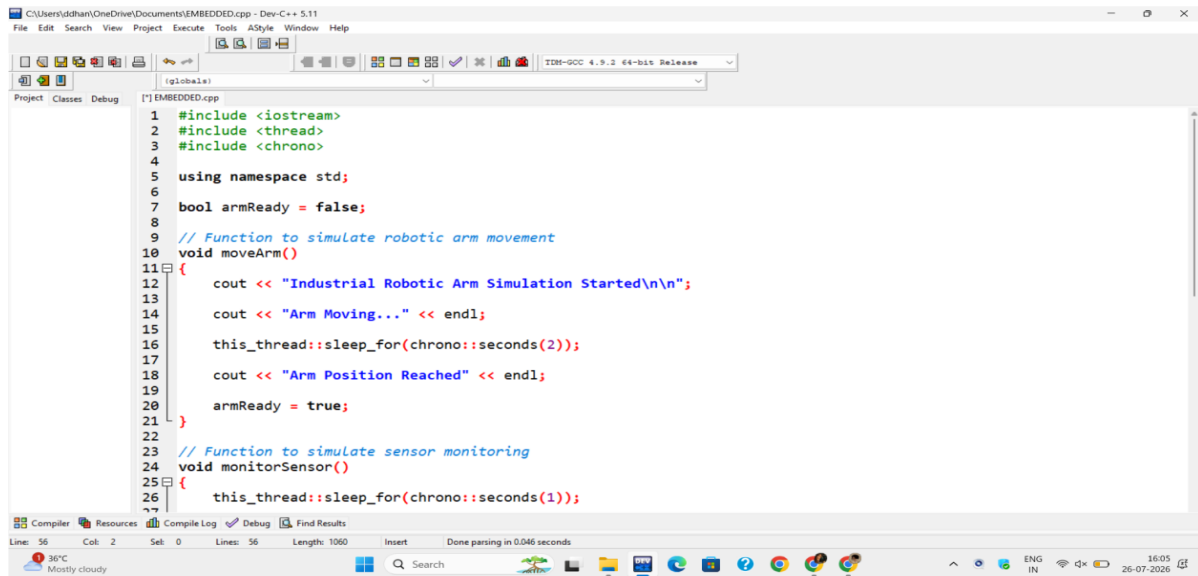


CO2-AT1 SLOT-C ECA1409 EMBEDDED SYSTEMS

Title-SystemC-Assisted Embedded C Simulation for an Industrial Robotic Arm with Concurrency and Synchronization

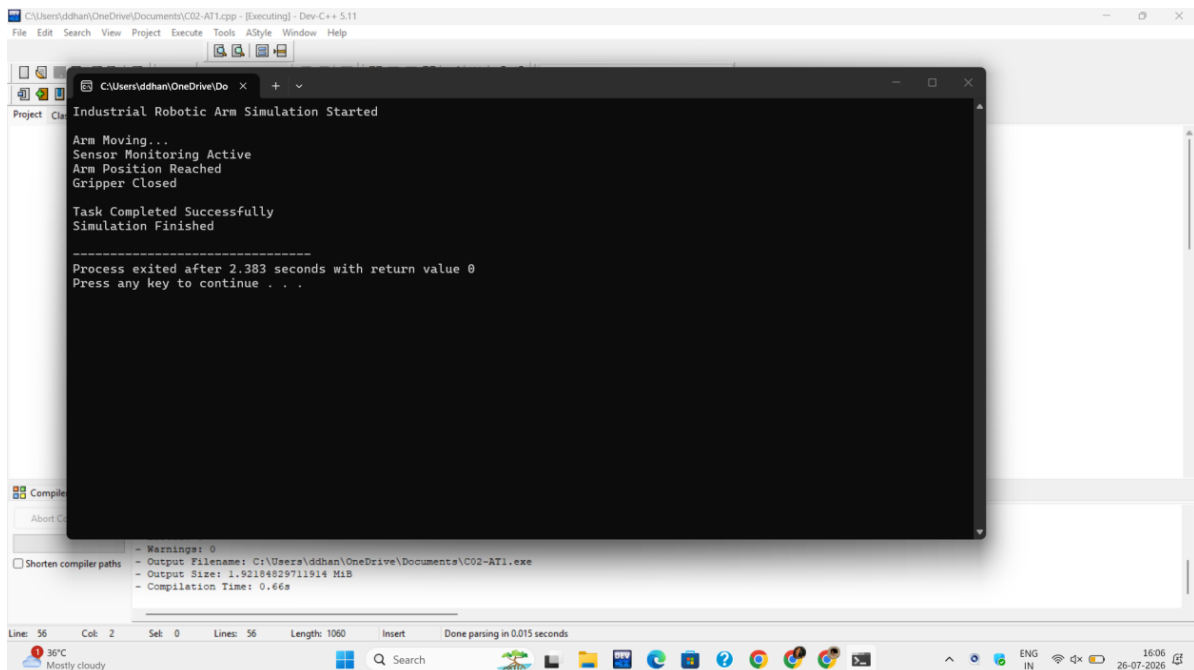
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OUTPUT



```
1 #include <iostream>
2 #include <thread>
3 #include <chrono>
4
5 using namespace std;
6
7 bool armReady = false;
8
9 // Function to simulate robotic arm movement
10 void moveArm()
11 {
12     cout << "Industrial Robotic Arm Simulation Started\n\n";
13     cout << "Arm Moving..." << endl;
14     this_thread::sleep_for(chrono::seconds(2));
15     cout << "Arm Position Reached" << endl;
16     armReady = true;
17 }
18
19 // Function to simulate sensor monitoring
20 void monitorSensor()
21 {
22     this_thread::sleep_for(chrono::seconds(1));
23 }
```

EMBEDDED C SAMPLE CODE AND SYSTEMC CODE IN DEV C++



```
Industrial Robotic Arm Simulation Started
Arm Moving...
Sensor Monitoring Active
Arm Position Reached
Gripper Closed
Task Completed Successfully
Simulation Finished
-----
Process exited after 2.383 seconds with return value 0
Press any key to continue . . .
```

OUTPUT BY MERGERING EMBEDDED C SAMPLE CODE AND SYSTEMC CODE